

Andrei-Eduard Cursaru

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WORK EXPERIENCE

Capillarity and rehology lab – Research Trainee

Feb 2024 - Present

Project leader dr. Claudiu Pătrașcu, POLITEHNICA București

- Fundamental and exploratory research activity in fluid mechanics.
- Acquired skills in high-speed imaging acquisition and conducted experimental studies of capillarity interfacial phenomena and Taylor flows, including numerical modeling of the aforementioned.

TU Darmstadt – Student Research Assistant

Jun 2025 - Sep 2025

Institute of Nano- and Microfluidics, Darmstadt

- Erasmus+ traineeship focused on developing and validating multiphase flow simulations using the software OpenFOAM (phase-field methods)
- Strong skills in computational fluid dynamics (CFD), numerical modeling, and scientific analysis
- Experience with programming, debugging, and optimizing simulation environments. Independent problem-solving and ability to manage complex technical tasks
- Effective teamwork through weekly research meetings and continuous collaboration with supervisors and colleagues
- Skills applicable to biomedical engineering, including modeling biological fluid behavior and supporting medical device development

EDUCATION

Bachelor's degree in Biomedical Engineering

2022 - present

Facultatea de Inginerie Medicală, UNSTPB

Expanded knowledge in STEM fields, with a focus on applying natural sciences such as biology and chemistry. Gained experience working with biological samples, cell cultures, and developing biocompatible materials, as well as testing their properties. Strengthened computer science skills through hands-on projects, including database creation and management using SQL and Python. Further applied physics principles in handling and testing small electrical circuits

EXTRA CURRICULAR ACTIVITIES

EELISA Scientific Student Competition, Honorable Mention

Madrid, 2026

Co-authored and presented the paper "Getting Taylor bubbles to move" in collaboration with Andra-Eleonora Rădan. Project supervised and coordonated by Conf. dr. ing. Claudiu Pătrașcu

Student Scientific Communication Session, 1st Prize

București, 2025

Co-authored and presented the paper "Mișcări atipice ale cavităților Taylor" in collaboration with Andra-Eleonora Rădan. Project supervised and coordonated by Conf. dr. ing. Claudiu Pătrașcu

Volunteerd at LSAC Students' Association

București, 2022 - 2024

Part of the Fundraising Team for MOFT II. Responsible for setting goals, organizing charitable events, and engaging consistently with sponsors, stakeholders, and team members to negotiate and maintain partnerships.

PROJECTS

Numerical simulations of multi-phase capillary flows	openFOAM
Fine-tuned image processing for detection of kidney stones	Python, ImageJ
Prototype for an automated database	Python, SQL
Medical research blog website	HTML, CSS
Model servo-articulated robotic hand	Arduino
8-bit CPU design and implementation from reference documentation	Digital

SKILLS

General	Computational Fluid Dynamics (openFOAM); AutoCAD; Rheological tests; High-speed image aquisition and processing; Biomaterials study and synthesis; Cell culture handling; Advanced material characterization techniques; Communication and teamwork
Programming	Python, Linux (Intermediate); C/C++ (Basic); HTML+CSS
Digital Competence	Microsoft Office , Database Management (SQL), MATLAB, OriginPro
Languages	Romanian (Native) ; English (Proficient) ; German (Elementary)
Other	Driving licence (AM,B)